

Design and Optimization of Machine Learning Models

Dados e Aprendizagem Automática

Luís Tiago Pereira Borges
PG55977
Universidade do Minho

Miguel Jacinto Dias Carvalho
PG57590
Universidade do Minho

Ricardo Filipe Araújo Pereira
PG56001
Universidade do Minho

I. INTRODUCTION

Cognitive decline, particularly in the form of conditions such as Mild Cognitive Impairment (MCI) and Alzheimer's Disease (AD), is a growing global health concern. These conditions affect the quality of life of patients and their families. MCI is an intermediate stage between normal aging and dementia, where individuals experience measurable cognitive decline without severe interference in daily activities. However, individuals with MCI, especially those with memory-related symptoms, face a high risk of progressing to AD.

Alzheimer's Disease is a neurodegenerative condition characterized by a gradual decline in memory and cognitive abilities. It causes a significant burden due to its increasing prevalence, high treatment costs, and severe impact on patients and caregivers. Early prediction of MCI progression to AD is critical for timely interventions that could slow or prevent further decline.

Advancements in medical imaging and data analysis provide opportunities to handle this problem. Magnetic Resonance Imaging (MRI) is a key tool for studying structural changes in the brain associated with cognitive decline, such as hippocampal atrophy and ventricular enlargement. Radiomics, a technique that extracts quantitative features from medical images, enables detailed analysis of these structural changes. By applying machine learning models to radiomic features, it is possible to predict cognitive transitions, offering valuable insights for clinicians and researchers.

This project focuses on analyzing radiomic features extracted from MRI scans of the hippocampus to predict whether individuals with MCI will progress to AD. The study uses datasets provided by the Alzheimer's Disease Neuroimaging Initiative (ADNI), preprocessed with the PyRadiomics library. The hippocampus dataset, chosen for its relevance to Alzheimer's research, is complemented by a control dataset from the occipital lobe, which is not typically associated with dementia.

The work includes two main tasks: dataset exploration, analysis, and preparation, followed by the design and optimization of machine learning models for classification. A Kaggle competition is used as a platform to evaluate the performance of these models based on the F1 macro score.

This report details the methods and results of these tasks, aiming to contribute to the development of reliable tools for predicting cognitive decline.

II. CRISP-DM METHODOLOGY

To structure the development of this project, we adopted the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology. The methodology is separated in six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Below, we outline how each phase was applied to this project:

1. Business Understanding

The primary objective of this project is to predict the progression of individuals with Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD) using radiomic features extracted from MRI scans. This predictive tool could help clinicians intervene earlier and more effectively in managing cognitive decline.

Two datasets were utilized: one focused on the hippocampus, a brain region central to Alzheimer's research, and a control dataset based on the occipital lobe. The project also includes a Kaggle competition to benchmark model performance using the F1 macro score.

2. Data Understanding

The datasets were analyzed to identify their structure, content, and quality. Key characteristics such as feature distributions, missing values, and class imbalances were examined.

The datasets include radiomic features extracted from MRI scans, such as shape, texture, and intensity-based metrics. Additionally, we explored the relationships between features and the target variable, *Transition*, which represents different stages of cognitive decline.

3. Data Preparation

This phase involved several preprocessing steps to ensure the data was suitable for machine learning. We experimented with more than a dozen different steps and various variations on each. The techniques used will be explained further, with more detail.

4. Modeling

Various machine learning models, including Random Forest, XGBoost, Logistic Regression, and Artificial Neural Networks, were developed and optimized. Hyperparameter tuning and cross-validation were employed to maximize performance, with a strong emphasis on the F1 macro score as the primary evaluation metric.

5. Evaluation

The performance of each model was evaluated using metrics such as the F1 macro score and confusion matrices.

6. Deployment

While deployment was outside the direct scope of this project, the Kaggle competition served as a platform to simulate real-world application scenarios. This step demonstrated the ability of the models to generalize to unseen data and provided additional validation of the methodology and results.

III. DATASET EXPLORATION AND DATA TREATMENT

This is the first step of the project and involves the first three phases of the CRISP-DM process. Below, we can find a precise description of each stage.

1. Dataset Exploration

The first step in the code consists of analyzing the train dataset. Here, we analyzed:

- Dataset Shape: Number of lines and columns
- Dataset Type: Data types present on the dataset
- Object Columns: Names of the columns of the type 'Object'
- Missing Values: Number of missing values
- Transition Column: Number of cases of each Transition type
- Unique value column: Number of columns filled with a unique value

This initial exploration is huge to guide us through the initial steps.

On a later phase, some more exploration was conducted. The Age and Sex features were analyzed, as well as the relationship between the Age and Transition features.

2. Removing Columns with irrelevant information

In this step, columns with the 'Object' value were evaluated. From these, the ones where all elements were unique were deleted, as they did not represent useful information (they were, mostly, identification columns).

However, three special columns were kept, as they had possibly valuable information but needed some data treatment. These columns were:

- diagnostics_Mask-original_BoundingBox
- diagnostics_Mask-original_CenterOfMassIndex
- diagnostics_Mask-original_CenterOfMass

From these three, we were able to conclude that diagnostics_Mask-original_CenterOfMassIndex and diagnostics_Mask-original_CenterOfMass had the same information, so we dropped the latter.

Also, later, constant features - features that contain the same value for all their rows - were dropped. These ended up providing no valuable information, so there was no problem eliminating them.

At an even later point, there were still useless features. These were the groups of features that had a correlation value of 1, which meant that they represented the same information and would not help a model to perform better. From these groups, all but one of them were dropped.

3. Data Engineering

In this section, we handled the two columns left from above. For the first feature, diagnostics_Mask-original_BoundingBox, we believe its format represents the 3 box coordinates (x,y,z) and its width, height and depth. So, columns referring to these features were created, and the extracted values were assigned to them. A column with the volume of the box was also added, calculated using the width, height and depth values.

A similar strategy was used on the diagnostics_Mask-original_CenterOfMassIndex feature, as the values here represent the x,y and z values for the center of mass. So, these features were added, as well as a 'distance' feature, which represents the distance from the center of mass to the bounding box.

4. Encoding

This quick step consisted of the encoding of the target values into numeric ones, so that the models would be able to handle them.

5. Outlier Treatment

Outlier treatment is a very important step in every machine learning project, and this one was no exception. Different values were tried for the upper and lower bounds, and the outliers, i.e., values outside of these, were replaced with the values corresponding to the closest bound.

6. Standardizing Data

Data standardization is also a major step. For it to be done, a scaler was created and applied to the training and testing dataset.

7. Oversampling

It was concluded, at the start, that the dataset was extremely unbalanced. To combat this, an oversampling technique was applied. After several tests, the chosen strategy gave 96 elements to each class but one, which ended up with only 31 elements. It was decided to not use 96 elements for all classes as there was one which, originally, had only 10 elements, and getting it to 96 would mean there was a class with 89.5% of its elements generated artificially, which is not recommended.

8. Feature Selection and Dimensionality Reduction

Since some models struggle with datasets with a large amount of features, it was decided to apply feature selection. Using the ANOVA technique, the number of features was reduced to 200.

It was also concluded that it would be beneficial to use a dimensionality reduction technique, which led the group to use the PCA tool to reduce the number of features to 150.

9. Importance Checking and Saving

At the end, a feature importance check was conducted, revealing the 20 most important features. After that, the new, treated datasets were saved, as they were ready to be applied to the models.

IV. MACHINE LEARNING MODELS

With the dataset ready, it was time to apply the models. During the process of development, the group tested with all sorts of models, including:

- Random Forest
- SVM
- XGBoost
- Logistic Regression
- LightGBM
- Artificial Neural Networks

It is also worth mentioning that some of those were combined into ensemble models with the development of stacking and voting versions.

Here's a concise description of each model:

A. Random Forest

This model constructs multiple decision trees during the training and combines the outputs into the result.

B. SVM

SVM is a model that works by finding the optimal hyperplane to separate data into different classes. It tries to maximize the margin between the closest data points of different classes, which are called the support vectors.

C. XGBoost

XGBoost is a gradient-boosting algorithm able to build sequential decision trees, where each tree corrects the errors of the one before.

D. Logistic Regression

Logistic Regression models predict the probability of an outcome belonging to a particular class by using a logistic function.

E. LightGBM

LightGBM is also a gradient-boosting algorithm, but it is specially designed for large datasets.

F. Artificial Neural Networks

These are computation models inspired by the human brain, as they consist of layers of nodes connected to each other.

Our approach to each model

For these models, the strategy applied was always the same: splitting the data into oversampled and non-oversampled, training with the complete data using cross validation, and calculating the scores using only the non-oversampled data, as using oversampling data on testing is not recommended.

After that, a confusion matrix is plotted, providing a visual representation of the results.

Chosen Model

The chosen model was an ANN with 7 layers. A visual representation of this network is provided below:

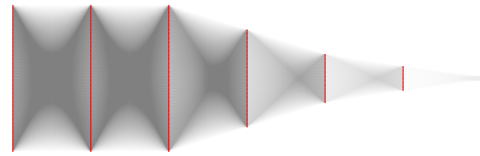


Fig. 1. Artificial Neural Network

This Artificial Neural Network starts with 3 layers of 150 nodes, which gradually reduce to 100, 50, 25 and, finally, 5, which is the number of possible values to predict.

Hyperparameter wise, it uses a dropout value of 0.5, a batch size value of 32, 75 epochs and a learning rate of 0.001.

This model was chosen because, from the consistent models, i.e., the models whose score was consistent locally and on kaggle, on the public dataset, this was the best. It presented a f1 macro score of 0.452 locally and 0.456 on the kaggle competition when tested on the public dataset. Having in mind that the group used cross-validation to get the local score, we were pretty confident that we had a good model in our hands.

However, the model scored only 0.366 on the private dataset, which means it was overfitting the data.

V. CRITICAL ANALYSIS

Besides analyzing the selected model, the group wants to use this section to reflect on the results as a whole. During the months of the competition, we submitted 196 entries, of which we have a complete record of about 140. These records give us the necessary information to reach some important and interesting conclusions.

Some relevant statistics are presented in the above graphic, which we will further analyze. This graphic shows the average values, for each mode, for local score, as well as the public and private datasets score and the difference between the latter two.

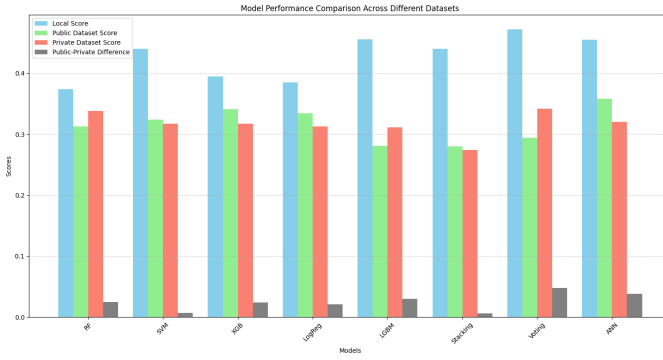


Fig. 2. Models' Statistics

The graph shows us that the SVM, LGBM, stacking, voting and ANN models were the best locally, all performing, on average, at above 0.40.

However, we can see the results being more consistent on the other 3:RF, XGB and LogReg, although this may be because the local results are inferior.

Looking at the difference between the public and private scores, it's interesting to note that this value is the lowest for the SVM and stacking models. This would indicate that these models are very reliable and their results on a smaller test dataset should be close to the ones on a bigger one.

Although these statistics are factual, they might not provide the best information. Since the group was trying to get the best possible model, some types of models were tested way more often than others, so these results don't necessarily reflect the effectiveness of the models in general, or their consistency.

It's also worth it spending a couple words on the best models developed. Now that there's access to the public dataset, we can list these three as our best models.

A. Random Forest Model (v27)

One of the earliest models we submitted was a simple Random Forest model, with 100 estimators. This model provided a private score of 0.392, very consistent with the local score of 0.399.

B. Artificial Neural Network (v105)

There was also an ANN which is a decent candidate for the best model. It was built with 6 layers and uses 75 epochs, providing a private score of 0.382. However, the local score is 0.504, showing close to no consistency.

C. Artificial Neural Network (v118)

This neural network uses 7 layers and 75 epochs, while also using a weight decay technique. Besides presenting a satisfying private score of 0.384, it also, like the one above, shows a great degree of inconsistency, providing a 0.526 local f1 macro score.

VI. CONCLUSION AND FUTURE WORK

By reading the critical analysis section, it's obvious the results were not what the group expected. Most of the models, including the selected one, overfitted the data, leading to a great fall in the competition ranking.

However, it is important to note that doesn't take away from all we learned during the development of this project. The challenges of this particular dataset encouraged us to learn about feature engineering, oversampling, feature selection, dimensionality reduction and various types of models and their hyperparameters. We also take away valuable lessons about overfitting and hope to learn better ways to identify it moving forward.